Lab Final

CSE 4308

DATABASE MANAGEMENT SYSTEMS LAB

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1 Lab Final

1. List the `customer_id`, `first_name`, and `last_name` of all customers whose `last_name` starts with the letter `'S'`. Order by `customer_id`.

2. Find the films that have been rented **more than 20 times**. Show film id, title, and rental count. Order by rental count descending.

3. Find the **top 10 films** by total revenue (sum of payment.amount). Show film id, title, and total revenue.

4. Show each staff member and the **number of rentals they handled**. Order by rental count descending.

5. Write an anonymous block that prints all customers from store 1 whose `active` status is 0. Print customer id and full name using a loop.

6. Write an anonymous block that takes a film id (e.g., 1) and prints: (i) the film title, (ii) the number of times it was rented, and (iii) the total revenue from that film.

7. Write an anonymous block that prints the **top 5 customers** by total spending. Print customer id, full name, and total spending using a loop.

8. Create a function that takes a film ID and a store ID and returns how many copies of that film exist in that store and how many are currently available.

9. Create a function `film_revenue_range` that returns the total revenue earned by a given film within a given date range.

10. Create a procedure that takes a rental ID and sets `return_date = NOW()` for that rental only if return_date is currently NULL. If it is already returned, raise a notice and do nothing. If rental id does not exist, raise an exception.

2 ER Diagram


actor
actor_id INT * pk
first_name VARCHAR(45) *
last_name VARCHAR(45) *
last_update DATETIME *
language
language_id INT * pk
name CHAR(20) *
last_update DATETIME *
film
film_id INT * pk
title VARCHAR(255) *
description TEXT
release_year YEAR
language_id INT *
rental_duration SMALLINT *
rental_rate DECIMAL(4,2) *
length SMALLINT
replacement_cost DECIMAL(5,2) *
rating film_rating_enum
last_update DATETIME *
special_features TEXT
fulltext_col TEXT *
category
category_id INT * pk
name VARCHAR(25) *
last_update DATETIME *
country
country_id INT * pk
country VARCHAR(50) *
last_update DATETIME *
city
city_id INT * pk
city VARCHAR(50) *
country_id INT *
last_update DATETIME *
address
address_id INT * pk
address VARCHAR(50) *
address2 VARCHAR(50)
district VARCHAR(20) *
city_id INT *
postal_code VARCHAR(10)
phone VARCHAR(20) *
last_update DATETIME *
film_actor
actor_id INT *
film_id INT *
last_update DATETIME *
inventory
inventory_id INT * pk
film_id INT *
store_id INT *
last_update DATETIME *
film_category
film_id INT *
category_id INT *
last_update DATETIME *
customer
customer_id INT * pk
store_id INT *
first_name VARCHAR(45) *
last_name VARCHAR(45) *
email VARCHAR(50)
address_id INT *
activebool BOOLEAN *
create_date DATE *
last_update DATETIME
active INT
staff
staff_id INT * pk
first_name VARCHAR(45) *
last_name VARCHAR(45) *
address_id INT *
email VARCHAR(50)
store_id INT *
active BOOLEAN *
username VARCHAR(16) *
password VARCHAR(40)
last_update DATETIME *
picture BLOB
rental
rental_id INT * pk
rental_date DATETIME *
inventory_id INT *
customer_id INT *
return_date DATETIME
staff_id INT *
last_update DATETIME *
store
store_id INT * pk
manager_staff_id INT *
address_id INT *
last_update DATETIME *
payment
payment_id INT * pk
customer_id INT *
staff_id INT *
rental_id INT
amount DECIMAL(5,2) *
payment_date DATETIME *